

On the Challenges of Unsupervised Speech Recognition in Tonal Languages and Semitic Languages

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Unsupervised speech recognition is a promising approach for democratising speech technologies because it can produce speech recognition models for new languages using only unpaired speech and text data. One approach for unsupervised speech recognition is the speech decipherment method [1] that uses a multilingual phone recogniser (MPR) to decode unpaired speech into phone sequences that are then deciphered into text in the target language, using an approach similar to deciphering substitution ciphers. While the method showed promising results on seven test languages [1], we have found that it can fail for languages that use lexical tone, vowel length, or abjad or abugida writing systems. In this work, we explore augmenting the phone sequences with suprasegmental information to enable unsupervised speech recognition for tonal languages and Semitic languages which use the abjad writing system.

Phonetic transcription: Phonetic transcriptions generally fall into two categories [2]. *Broad* transcriptions have minimal suprasegmental detail such as intonation, stress or vowel length. For languages which require this detail, there may be arbitrarily more homophones which makes speech decipherment difficult. *Narrow* transcriptions aim to include as much suprasegmental detail as possible. Thus, our work uses unsupervised methods to augment the MPR transcriptions with suprasegmental detail related to vowel length and lexical tone.

Tonal Languages and Abjad Languages: Tonal languages use variations in pitch to lexically disambiguate words. Additionally, many abjad languages lexically distinguish between short vowels and long vowels. Abjad scripts explicitly write consonants, and (short) vowels are inferred based on the surrounding context. For such languages, the lack of tone or vowel length information in MPR decoded phone sequences makes speech decipherment difficult. Additionally, native speakers of both tonal languages and abjad languages may omit diacritics or letters related to tone and vowels in written texts, and infer the meaning of words from surrounding context [3, 4]. This conflicts with prior research which suggests that diacritics are necessary for good performance of language models [5]. We find that this missing lexical information in written texts, used to train the language model for speech decipherment, also makes speech decipherment difficult. Therefore, we use clustering techniques to restore vowel length and lexical tone information lost during MPR decoding, using audio of the target language.

Method: To train the clustering models used to augment MPR decoded phone sequences, we perform forced alignment using Kaldi [6] to derive audio segments for each phone from the training corpus. Given these audio segments, feature vectors are derived for both vowel length modelling and lexical tone

modelling.

For vowel length modelling, we use a two-component Gaussian Mixture Model (GMM) to cluster each vowel phone into *long* or *short* implicit classes. The GMM is trained on two-dimensional feature vectors: the raw and log vowel duration. We use log vowel duration features as it corrects the observed left-skewed distribution of the vowels. During inference, the model augments each vowel phone with either implicit class.

For lexical tone modelling, we pre-process each vowel phone audio segment by extracting the log F_0 contour and applying per-speaker z -score normalisation to remove speaker-specific information. We obtain feature vectors by subsequently applying Discrete Cosine Transform (DCT); truncating the output vectors to length 5, and removing c_0 to further remove speaker-specific information. K -means is used to cluster the lexical tone representations into M tonal classes.

Results and Community Contribution: Preliminary results highlight that adding vowel length information to MPR decoded phones results in sharp performance improvements for Hebrew, where the Word Error Rate (WER) falls from **107.7** to **70.7**, which is sufficient to bootstrap incremental semi-supervised training that can further improve the WER. We encourage researchers working on unsupervised speech recognition to test their methods on tonal languages and languages with abjad or abugida writing systems, to help ensure that speech technologies are also available for these languages.

Generative AI Statement

No generative AI tools were used during this project.

1. References

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